

Barycentric Compression Witnessing

A Datum-Ledger Method for Group-Context Parity in Declared Weak-Field Scenes

Working Replacement for GCS Paper VI / GFRO

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Core doctrine. Context is not emitted by pairwise equality. Context is declared, compressed, tested, ledgered, and audited.

Abstract

Declared weak-field gravitational scenes require more than a physical source roster. They also require rules for how resolved child structure is carried, compressed, compared, and handed off to an enclosing parent context. Earlier Atlas work used scalar source-readable parity branches, including pairwise quantities proportional to M_i/r_i^3 , to recover equality surfaces and many-source construction scaffolds. Those pairwise branches remain useful as drafting and indexing tools, but they do not decide when a resolved group can be represented by a parent-readable effective object.

This paper introduces Barycentric Compression Witnessing (BCW), a datum-ledger method for testing group-context compressibility in declared weak-field scenes. Given a declared source group G , BCW compares the resolved electric-Weyl/tidal field of the group against a declared surrogate field, beginning with the barycenter monopole. The tensor residual

$$\Delta E_G^{(L)}(x) = E_{\text{actual},G}(x) - E_{\text{sur},G}^{(L)}(x) \quad (1)$$

measures failure of group compressibility at x . The normalized witness

$$\chi_G^{(L)}(x) = \frac{\left\| \Delta E_G^{(L)}(x) \right\|_F}{\left\| E_{\text{sur},G}^{(L)}(x) \right\|_F + \epsilon_\chi} \quad (2)$$

defines candidate compression-parity surfaces such as $\chi_G^{(L)} = 1$. These surfaces are not physical membranes, force cutoffs, horizons, or modifications of general relativity. They are diagnostic bookkeeping structures indicating where unresolved internal group structure has tensor weight comparable to the declared parent-readable surrogate.

BCW is governed by datum discipline: sources contribute, datums interrogate, witnesses report, ledgers remember, and rendering audits. Shell-local Cartesian cell complexes provide the primary sampling support, while Node-B omni-radial sampling may be toggled as a continuity and crossing audit. A first synthetic calibration run, BCW-L001, produced nontrivial

$\chi_G = 1$ structures in a twelve-source weak-field scene. These structures persisted under focused Cartesian refinement and were echoed by dense Node-B radial audit, while also revealing non-star-shaped, multi-crossing behavior that left topology uncertified. BCW therefore replaces the over-promoted GFRO field-emission framing with a narrower instrument: parity geometry is earned through declared compression residuals, ledgered sampling, and disciplined validation.

Notation

Symbol / Term	Meaning
S	Declared physical source roster
G	Declared child/group subset of S
Π	Declared parent context
x	Query point
p_i	Position of source i
M_i	Mass of source i
ϵ_i	Source-model softening length for source i
ϵ_B	Barycenter-surrogate softening length
ϵ_χ	Numerical denominator floor in χ
$E_i(x)$	Weak-field electric-Weyl/tidal contribution of source i
$E_{\text{actual},G}(x)$	Resolved tensor field of group G
M_G	Total mass of group G
B_G / Node B	Mass barycenter / compression witness
C_G / Node G	Geometric centroid / morphology witness
$E_{\text{bar},G}(x)$	Barycenter monopole surrogate, $L = 0$
$E_{\text{sur},G}^{(L)}(x)$	Declared surrogate field of group G at rung L
$\Delta E_G^{(L)}(x)$	BCW tensor residual
$\chi_G^{(L)}(x)$	Self-compression witness
$\chi_{G \Pi}^{(L)}(x)$	Parent-weighted compression witness
$\Sigma_G^{(L)}$	Candidate BCW parity surface, $\chi_G^{(L)} = 1$
D_Σ	Finite sampling/support band around Σ
Q_i	Legacy scalar source-readable scaffold contribution
Ψ_{ij}	Pairwise scalar residual $Q_i - Q_j$
Σ_{ij}	Pairwise parity scaffold surface
Node 0	Global registration datum
Node 1	Lawful parent scene relation
Node Ψ	Extracted diagnostic layer

1 Introduction: The Context Problem After Pairwise Parity

Atlas began with a productive observation: in a declared weak-field multi-source scene, there are locations where one source-readable contribution equals another. In the scalar point-source branch, a readable contribution such as

$$Q_i(x) \propto \frac{M_i}{r_i(x)^3} \quad (3)$$

supports pairwise residuals

$$\Psi_{ij}(x) = Q_i(x) - Q_j(x) \quad (4)$$

and equality sets

$$\Sigma_{ij} = \{x : \Psi_{ij}(x) = 0\}. \quad (5)$$

In narrow point-source cases, these equality sets recover Apollonius-type construction surfaces. In many-source scenes, they can assemble into faces, seams, junctions, and candidate parity networks. This branch was useful because it gave the lab geometry it could see and extract. It provided construction lines through the fog.

But pairwise equality is not the same as context.

A declared group can contain many internal source-source contests while still being readable from far away as a single effective mass. Conversely, a group can fail to compress into a simple parent-readable object in regions where no single pairwise surface tells the full story. A source-source equality surface asks whether two isolated contributions balance under a chosen scalar rule. The context problem asks when a resolved child or group must remain resolved, and when it may be compressed for a parent scene.

This is the gap the prior GFRO framing attempted to cross. GFRO treated source-context residual construction and coordinate emission as if parity geometry could be constructed directly from declared residual equations. That was a useful discovery path, but the claim was over-promoted. The field-emission language made the method sound more complete than it was. It risked treating a drafting procedure as a context calculus.

The corrected question is not: can a residual equation emit the parity network? The corrected question is: given a declared child/group and a declared parent-readable surrogate, where does the resolved group field fail to compress?

BCW answers that narrower question. It does not replace general relativity. It does not introduce a new source or new force. It does not turn diagnostic surfaces into physical boundaries. It provides a way to measure compression failure inside a declared weak-field time slice.

This reframes Paper VI. The replacement is not a better emitter. It is a better witness.

Figure 0 placeholder. Minimal BCW cartoon. A declared group of physical sources is compared with a ghost barycenter monopole surrogate at Node B. The figure should show the resolved group field, the surrogate field, the residual, and the candidate $\chi = 1$ support. Node B must be labeled as a compression witness, not a source.

Figure 1: Minimal BCW cartoon.

2 Correction Note: GFRO as Demoted Lineage

The prior Paper VI / GFRO manuscript stabilized around the Gravitational Field-Relation Operator, with the field-sieve procedure described as a coordinate-emission mechanism. Its public ambition was direct construction of gravitational parity networks from source-context residuals.

The corrected status is more conservative.

GFRO remains historically important because it exposed a valuable pressure point: the method needed field-derived residuals rather than only scalar pairwise geometry. It pushed Atlas away from flat pairwise accumulation and toward source-context comparison. It also forced the lab to confront tensor witnesses, residual cleanliness, entropy consistency, eigenframe handoff, and retention discipline.

But GFRO should no longer be treated as the default engine doctrine.

The demotion is specific:

- The full-context field-emission framing is demoted.
- The idea that residual emission alone lawfully produces full context parity geometry is not certified.
- Pairwise and GFRO-style residual branches remain useful as scaffolds, indexes, and historical lineage.
- Serious group-context parity requires datum-ledger evaluation of resolved fields against declared surrogates.

The compact correction is:

Pairwise parity drafts seams. BCW tests context. The ledger decides retention.

GFRO is therefore preserved as prior candidate lineage, not erased. Its value is that it revealed the need for a field-derived context witness. Its failure is that it promoted residual emission before the witness law had been declared.

3 Relation to the GCS Series

BCW is not a reset of the GCS program. It is a correction of the Paper VI branch after the first five public papers had already built useful pieces of the architecture.

3.1 GCS I: Contribution-Context Parity

GCS I defines the root object: a gravitational coherence surface is the parity surface where a declared gravitational contribution reaches equality with its derived gravitational context under a declared readability rule. It also makes the crucial non-claim: the GCS is not a force boundary, physical membrane, causal horizon, trapping horizon, or dynamical stability limit. This remains intact.

BCW preserves the GCS root definition but changes the serious group-context readability rule. Instead of treating scalar Q parity as sovereign for complex group-context scenes, BCW declares a tensor compression residual:

$$\Delta E_G^{(L)}(x) = E_{\text{actual},G}(x) - E_{\text{sur},G}^{(L)}(x). \quad (6)$$

Thus BCW is still a contribution-context parity branch. The contribution is now the resolved group field. The context is a declared parent-readable surrogate. The parity witness is the equality between the residual norm and the surrogate norm.

3.2 GCS II: Many-Source Pairwise Scaffold

GCS II develops many-source pairwise parity networks. It defines pairwise comparison fields and pairwise parity surfaces. This remains valid as the scaffold branch. BCW does not erase it. It prevents the scaffold from being mistaken for a full context calculus. Pairwise parity continues to draft candidate seams and to seed candidate cell complexes. BCW then tests whether those candidate regions matter to group-context compression.

3.3 GCS III: Mesoscale Interface Geometry

GCS III introduced the mesoscale comparison layer between pointwise local evaluation and coarse averaging. It argued that parity faces, seams, low-margin regions, and candidate transition loci can function as derived diagnostic interface geometry rather than as physical membranes.

BCW sharpens that mesoscale hypothesis. Instead of asking only whether source-readable pairwise surfaces persist, it asks whether a resolved group fails to compress into its parent-readable surrogate. This is a more direct test of the middle layer between full local resolution and coarse summary.

3.4 GCS IV: Extraction Survives, Sovereignty Narrows

GCS IV showed proof-of-concept extraction under the scalar branch $Q_i \propto M_i/r_i^3$. It established that recoverable boundary-support geometry can appear in declared weak-field scenes, while explicitly keeping the claim operational and branch-relative.

BCW treats GCS IV as valid but narrower. The extraction lesson survives: ledged support can be recovered and audited. The operator sovereignty changes: scalar extraction becomes a baseline scaffold, while serious group-context compression requires the BCW tensor residual branch.

3.5 GCS V: Nesting Handoff Becomes the Target

GCS V named the nesting domain handoff problem: high-resolution child scenes must be embedded within, supported by, and interpreted through lower-resolution parent contexts. It argued that GR supplies the local law, while a context calculus is needed to formalize how declared gravitational scenes nest, hand off, and preserve identity across scale.

BCW is the first concrete instrument aimed directly at that static-slice handoff problem. It does not solve dynamics, but it supplies a compression test: can the parent read this resolved child/group as a declared surrogate here, or must the child remain resolved?

3.6 Lineage Summary

Paper / Branch	Survives as	BCW correction
GCS I	Root parity definition	Tensor compression rule
GCS II	Pairwise scaffold	Not full context solution
GCS III	Mesoscale layer	Compression residual test
GCS IV	Scalar extraction proof	Operator sovereignty narrows
GCS V	Nesting handoff problem	BCW becomes first instrument
Information Boundaries	Attribution interpretation	BCW supports replace scalar supports
Three Measures	Budget / jurisdiction / compression	BCW bridges jurisdiction/compression

4 Relation to Information Boundaries

The Information Boundaries paper interprets a closed GCS as a modular attribution surface: a boundary where source-resolved gravitational identity becomes maximally contested by contextual explanation. In its minimal normalized support model, binary attribution entropy is maximized exactly at parity:

$$Q_A = Q_K. \quad (7)$$

BCW updates the support model without rejecting the interpretation.

Instead of treating the supports as scalar Q_A and Q_K in serious group-context scenes, BCW supplies tensor-derived supports:

$$R_G^{(L)}(x) = \left\| \Delta E_G^{(L)}(x) \right\|_F \quad (8)$$

and

$$S_G^{(L)}(x) = \left\| E_{\text{sur},G}^{(L)}(x) \right\|_F. \quad (9)$$

The normalized BCW attribution weights can then be written:

$$w_{\text{res}}(x) = \frac{R_G^{(L)}(x)}{R_G^{(L)}(x) + S_G^{(L)}(x)} \quad (10)$$

and

$$w_{\text{sur}}(x) = \frac{S_G^{(L)}(x)}{R_G^{(L)}(x) + S_G^{(L)}(x)}. \quad (11)$$

The binary attribution entropy is

$$S_{\text{attr}}(x) = -w_{\text{res}}(x) \log w_{\text{res}}(x) - w_{\text{sur}}(x) \log w_{\text{sur}}(x). \quad (12)$$

In this support model, entropy is maximized when $w_{\text{res}} = w_{\text{sur}}$, which is equivalent to

$$\left\| \Delta E_G^{(L)}(x) \right\|_F = \left\| E_{\text{sur},G}^{(L)}(x) \right\|_F. \quad (13)$$

Thus the BCW parity condition becomes the compression-attribution entropy ridge. The information-boundary interpretation naturally migrates from scalar contribution/context supports to BCW compression supports.

This entropy branch remains a consistency interpretation, not an independent proof. It says that at BCW parity, the model has equal support for two bookkeeping explanations: unresolved group detail must be carried, versus parent-readable surrogate remains adequate.

5 Relation to Three Measures

The Three Measures program separates gravitational identity into budget, jurisdiction, and compression.

BCW belongs most directly to the jurisdiction/compression interface. Budget asks what finite curvature or field budget the source/group carries. Jurisdiction asks where that contribution remains meaningfully attributable. Compression asks when resolved detail can be replaced by an effective parent-readable representation.

BCW provides an operational bridge:

- $\Delta E_G^{(L)}$ measures compression failure.
- $\chi_G^{(L)}$ marks where unresolved group detail has equal tensor weight to the declared surrogate.
- The surrogate ladder L asks how much internal structure the parent must retain before the group becomes compressible to tolerance.

In this sense, BCW-L0 is the monopole jurisdiction/compression test, while higher rungs $L \geq 2$ become compression-order tests. The Three Measures paper should eventually cite BCW as the replacement for GFRO-style jurisdiction claims.

6 Datum Discipline

BCW is not just a formula. It is a formula embedded inside a witness law.

The governing discipline is:

Sources contribute. Datums interrogate. Witnesses report. Ledgers remember. Rendering audits.

A physical source contributes to the modeled field. A datum registers, samples, launches, measures, brackets, or reports. A witness is a datum with a reporting role. A render is a downstream audit surface for human interpretation; it does not decide mathematical status.

The relevant node roles are:

- **Node 0:** global registration datum.
- **Node 1:** lawful parent scene relation.
- **Node B:** mass barycenter / compression witness. Not a source.
- **Node G:** geometric centroid / morphology witness. Not a source.
- **Node Ψ :** extracted diagnostic layer.

This node discipline matters because the GFRO failure was partly a role confusion. The method wanted a context object, but it blurred the distinction between source, surrogate, datum, and emitted geometry. BCW makes the distinction explicit. The barycenter surrogate is not a source; it is an evaluated parent-readable comparison model. The residual between the resolved group and that surrogate is not a new field; it is a diagnostic tensor difference.

7 Weak-Field Field Branch

This draft works in a declared weak-field, static or quasi-static time-slice approximation. The field object is the electric-Weyl / Newtonian tidal tensor branch.

For a pointlike source i of mass M_i at position p_i , evaluated at query point x , define

$$r_i = x - p_i, \quad (14)$$

$$d_i = \sqrt{\|r_i\|^2 + \epsilon_i^2}, \quad (15)$$

$$n_i = \frac{r_i}{d_i}. \quad (16)$$

The softened weak-field tidal/electric-Weyl contribution is

$$E_i(x) = \frac{GM_i}{d_i^3} (3n_i n_i^T - I). \quad (17)$$

For a declared group G , the resolved group field is

$$E_{\text{actual},G}(x) = \sum_{i \in G} E_i(x). \quad (18)$$

This resolved field is the actual side of the comparison. It is built from the physical source roster. Only the physical sources contribute.

The softening ϵ_i is a source-model softening parameter. It regularizes the declared point-source approximation near source locations and is part of the scene declaration. In production work, ϵ_i may be common across synthetic point sources or source-specific when physical size, numerical resolution, or mass scale requires it. Either choice must be declared.

8 The Barycenter Surrogate

Let the declared group G have total mass

$$M_G = \sum_{i \in G} M_i \quad (19)$$

and mass barycenter

$$B_G = \frac{\sum_{i \in G} M_i p_i}{\sum_{i \in G} M_i}. \quad (20)$$

The first parent-readable surrogate rung is the barycenter monopole. It asks whether the group can be read as one collapsed mass located at B_G .

For the barycenter surrogate,

$$r_B = x - B_G, \quad (21)$$

$$d_B = \sqrt{\|r_B\|^2 + \epsilon_B^2}, \quad (22)$$

$$n_B = \frac{r_B}{d_B}, \quad (23)$$

and

$$E_{\text{bar},G}(x) = \frac{GM_G}{d_B^3} (3n_B n_B^T - I). \quad (24)$$

$E_{\text{bar},G}$ is not an additional physical source. It is a declared comparison surrogate. It is how the parent might compress the child/group at monopole order.

Because Node B is not a source, ϵ_B has a different status from ϵ_i . It does not regularize a physical source model. It protects the non-physical surrogate evaluation from artificial singular behavior at a derived witness point. The barycenter no-claim zone marks the region where the surrogate may be numerically useful but interpretively restricted.

For BCW-L001, the adopted rule was

$$\epsilon_B = \max(\text{source_softening_scale}, \beta_B R_{\text{group}}, \text{minimum_numerical}). \quad (25)$$

The recovered values are:

$$\begin{aligned} \text{source_softening} &= 0.04863453476582329, \\ \beta_B &= 0.04, \\ R_{\text{group}} &= 1.3895581361663797, \\ \epsilon_B &= 0.05558232544665519, \\ \text{barycenter_surrogate_no_claim_radius} &= 0.11116465089331037. \end{aligned}$$

A future sensitivity audit must test whether $\chi = 1$ support geometry shifts materially under β_B variation.

9 The BCW Residual

The core BCW correction is the tensor residual

$$\Delta E_G^{(0)}(x) = E_{\text{actual},G}(x) - E_{\text{bar},G}(x) \quad (26)$$

for the monopole rung $L = 0$.

More generally,

$$\Delta E_G^{(L)}(x) = E_{\text{actual},G}(x) - E_{\text{sur},G}^{(L)}(x), \quad (27)$$

where $E_{\text{sur},G}^{(L)}$ is the declared surrogate at rung L .

This residual measures failure of group compressibility.

If $\Delta E_G^{(L)}$ is small relative to the surrogate, the parent-readable representation is adequate under the declared tolerance. If $\Delta E_G^{(L)}$ is large, the unresolved internal structure of the group matters and must be retained or represented with a higher surrogate rung.

The normalized witness is

$$\chi_G^{(L)}(x) = \frac{\left\| \Delta E_G^{(L)}(x) \right\|_F}{\left\| E_{\text{sur},G}^{(L)}(x) \right\|_F + \epsilon_\chi}. \quad (28)$$

At $L = 0$,

$$\chi_G^{(0)}(x) = \frac{\left\| \Delta E_G^{(0)}(x) \right\|_F}{\left\| E_{\text{bar},G}(x) \right\|_F + \epsilon_\chi}. \quad (29)$$

The denominator floor ϵ_χ is a numerical regularization parameter, not a physical scale. It prevents division instability where the surrogate norm is very small. It must be declared separately from ϵ_i and ϵ_B .

In BCW-L001, ϵ_χ was not stored in the scene passport. The reproduced refinement/radial audit evaluator used $\epsilon_\chi = 10^{-12}$ as a floating numerical guard. Before any public release, this should be confirmed against the original first-pass generation script if archived, or declared explicitly as the reproduced evaluator guard. An ϵ_χ sensitivity check remains required.

The candidate BCW compression-parity surface is

$$\Sigma_G^{(L)} = \{x : \chi_G^{(L)}(x) = 1\}. \quad (30)$$

Equivalently,

$$\Sigma_G^{(L)} = \left\{ x : \left\| \Delta E_G^{(L)}(x) \right\|_F = \left\| E_{\text{sur},G}^{(L)}(x) \right\|_F \right\}. \quad (31)$$

At $\chi_G^{(L)} = 1$, the unresolved internal structure of the group has equal Frobenius weight to the declared parent-readable surrogate.

That is not a physical boundary. It is a bookkeeping parity condition. It identifies where compression and resolution have equal claim under a declared witness.

10 Parent-Context Version

A true context scene contains both a resolved child/group G and a declared parent context Π .

Let the parent field be

$$E_{\Pi}(x) = \sum_{j \in \Pi} E_j(x). \quad (32)$$

The resolved full scene is

$$E_{\text{scene}}(x) = E_{\Pi}(x) + E_{\text{actual},G}(x). \quad (33)$$

The parent-compressed scene at surrogate rung L is

$$E_{\text{scene,compressed}}^{(L)}(x) = E_{\Pi}(x) + E_{\text{sur},G}^{(L)}(x). \quad (34)$$

The direct compression error is

$$E_{\text{scene}}(x) - E_{\text{scene,compressed}}^{(L)}(x) = E_{\text{actual},G}(x) - E_{\text{sur},G}^{(L)}(x) \quad (35)$$

$$= \Delta E_G^{(L)}(x). \quad (36)$$

This cancellation is important but limited. It holds for the self-compression question when the declared parent field Π is held fixed across the resolved and compressed comparisons. In that case, the child-compression residual isolates the child’s unresolved contribution.

However, “holding the parent fixed” is a modeling declaration, not a universal nesting solution. If the parent field also contains subgroups that should be compressed, or if several nested children are being compressed simultaneously, the cancellation above does not automatically solve the hierarchy. Multi-level nesting requires an iterative or hierarchical BCW construction, with separate group declarations, surrogate choices, and residual ledgers at each level.

A parent-weighted witness can therefore be defined:

$$\chi_{G|\Pi}^{(L)}(x) = \frac{\left\| \Delta E_G^{(L)}(x) \right\|_F}{\left\| E_{\Pi}(x) + E_{\text{sur},G}^{(L)}(x) \right\|_F + \epsilon_{\chi}}. \quad (37)$$

The self-compression witness χ_G asks: how large is the unresolved child detail compared to the child’s surrogate? The parent-weighted witness $\chi_{G|\Pi}$ asks: how large is the unresolved child detail compared to the parent-compressed scene?

BCW-L001 used $\chi_G^{(0)}$ to isolate the instrument. A true child-in-parent benchmark should compute both $\chi_G^{(0)}$ and $\chi_{G|\Pi}^{(0)}$.

11 The Surrogate Ladder

The barycenter monopole is the first surrogate rung, not the whole story.

At $L = 0$, the parent asks whether the group is readable as a single mass at its barycenter.

At higher surrogate rungs, the parent may carry more of the group’s internal structure:

- $L = 0$: monopole barycenter surrogate.
- $L = 2$: quadrupole-augmented surrogate.
- $L = 3$: octupole-augmented surrogate.
- Higher L : declared multipole or effective representation branches.

The surrogate ladder asks: how much internal structure must the parent carry before the group becomes compressible to tolerance?

The ladder is promissory in this draft. Only $L = 0$ has been implemented in BCW-L001. The $L = 2$ quadrupole surrogate is not a plug-in label. It requires a declared quadrupole moment tensor of the group relative to a declared origin, typically Node B, and that tensor is group-declaration-relative. Its evaluation must specify convention, trace treatment, orientation, softening/interior rule, and domain of validity. Therefore the $L = 2$ rung is a future method object, not a completed result.

12 Pairwise Parity as Scaffold

Pairwise parity remains valuable.

The corrected architecture does not throw away Q , Apollonius geometry, or pairwise seams. It demotes them to their lawful role.

The pairwise branch

$$Q_i(x) \propto \frac{M_i}{r_i(x)^3} \quad (38)$$

with

$$\Psi_{ij}(x) = Q_i(x) - Q_j(x) \quad (39)$$

and

$$\Sigma_{ij} = \{x : \Psi_{ij}(x) = 0\} \quad (40)$$

can identify where isolated source-readable contributions balance. These surfaces can seed candidate cell complexes, guide refinement, and preserve historical continuity with earlier GCS work.

But pairwise parity does not decide group-context compression. A pairwise seam may be geometrically real as a drafting surface while being irrelevant to the resolved group’s compression failure. Conversely, BCW may find compression-failure structures that do not reduce to one pairwise contest.

The relation is therefore:

Pairwise parity drafts seams. BCW tests group-context compressibility. Ledger and validation decide retention.

Or more compactly:

Apollonius draws construction lines. BCW decides whether the drawing belongs to the scene.

13 Sampling: Cartesian Cell Complexes and Radial Audit Toggles

BCW surfaces are mathematical sets. Their finite evaluation requires datum support.

The primary BCW sampling support is the shell-local Cartesian cell complex. “Cartesian” here does not mean a single rectangular box as the object. It means globally aligned cells selected around contested geometry. Each cell has the same coordinate orientation, even when local resolution, density, and support shape vary.

The BCW surface remains the mathematical object:

$$\Sigma_G^{(L)} = \{x : \chi_G^{(L)}(x) = 1\}. \quad (41)$$

The sampling band is the witness support:

$$D_\Sigma = \{C_k : C_k \text{ intersects the declared near-parity band}\}. \quad (42)$$

The rendered geometry is downstream: crossing cells, near-band cells, overlap cells, low-margin regions, seams, component candidates, radial audit points, and retained or demoted structures.

The support band is not the surface. It is the finite witness structure that lets the method bracket, refine, overlap, and audit the surface.

BCW also supports an optional omni-radial audit toggle. For the barycenter compression branch, the natural radial origin is Node B:

$$x(r, n) = B_G + rn. \quad (43)$$

The radial audit searches for

$$\chi_G(B_G + rn) = 1 \quad (44)$$

along declared directions n and radial samples r .

The radial audit reports no-crossing rays, single-crossing rays, multi-crossing rays, crossing-radius distributions, and, in later implementation, nearest Cartesian cell distances.

It does not replace Cartesian extraction. It is a continuity and crossing witness. This distinction matters because BCW-L001 showed that the compression-delta object may be non-star-shaped from Node B, with no-crossing and multi-crossing rays. Radial sampling can fill in visual continuity, but it should not be forced into a one-radius-per-direction shell model.

14 Topology Certification Gap

BCW-L001 showed persistence of $\chi = 1$ support under focused refinement, but it did not certify topology.

Topology certification would require more than visually coherent lobes. At minimum, a topology claim requires one of the following:

1. Analytic continuation or implicit-surface arguments showing that the $\chi = 1$ set has a stable local/global connectivity structure under the declared branch.
2. Convergent mesh-based surface reconstruction across refinement levels, with stability of connected components, holes, tunnels, and boundary behavior.
3. Cross-witness reconciliation showing that Cartesian cell complexes, radial audits, and any contour-following or marching reconstruction agree within declared tolerance.
4. Sensitivity sweeps over resolution, near-band width, ϵ_χ , ϵ_B , source softening, and Node 0 registration.

BCW-L001 attempted none of these as certification. It established persistence under focused refinement and cross-witness echo under radial audit. Therefore the correct claim is:

BCW compression-delta parity persisted in the sandbox. Topology remains uncertified.

15 BCW-L001: Synthetic Calibration

BCW-L001 was designed as a sandbox methodology test, not as public evidence. Its job was to determine whether the barycenter compression residual produced nontrivial structure at all.

The scene used twelve softened point sources in a non-symmetric bounded 3D arrangement.

source_id	mass	x	y	z
S00	30.0	0.00	0.00	0.00
S01	18.0	0.78	0.18	-0.06
S02	12.0	-0.48	0.64	0.10
S03	8.0	-0.68	-0.44	-0.05
S04	5.0	0.32	-0.72	0.22
S05	3.0	1.05	-0.36	0.16
S06	2.0	-1.02	0.08	-0.18
S07	1.5	0.18	1.04	-0.12
S08	1.2	-0.22	-1.08	0.07
S09	1.0	0.92	0.72	0.24
S10	0.8	-0.88	0.86	-0.26
S11	0.6	0.52	-1.18	-0.22

The declared group was G_{all} , all twelve sources.

Recovered node values:

$$\begin{aligned}\text{Node0} &= [0.0, 0.0, 0.0], \\ \text{NodeB} &= [0.2066345123258521, 0.07331806615776081, 0.019676978637383485], \\ \text{NodeG} &= [0.04083333333333334, -0.021666666666666657, -0.00833333333333331].\end{aligned}$$

Node 0 was the neutral global origin. Node B was the mass barycenter of G_{all} . Node G was the geometric centroid. The branch was weak-field electric-Weyl / tidal tensor. The surrogate rung was $L = 0$ barycenter monopole. The run used normalized toy units with $G = 1$.

Recovered numerical hygiene values:

$$\begin{aligned}\text{source_softening} &= 0.04863453476582329, \\ \epsilon_B &= 0.05558232544665519, \\ \beta_B &= 0.04, \\ \text{barycenter_surrogate_no_claim_radius} &= 0.11116465089331037, \\ R_{\text{group}} &= 1.3895581361663797.\end{aligned}$$

ϵ_χ was not stored in the passport. The reproduced evaluator path used $\epsilon_\chi = 10^{-12}$; this remains a confirmation item before public release.

BCW-L001 is paired with the support note:

`BCW_L001_synthetic_12_source_run_note_v0.2_exact_metadata_fill.txt`

15.1 Sampling Results

First scout:

- Grid: $46^3 = 97,336$ points.
- Broad band: $0.5 \leq \chi \leq 2.0$.
- Band points: 7,442.
- Initial $\chi = 1$ crossing cells: 2,618.
- Initial shell-local cell complex: 7,676 cells.

Representative refinement scout:

- Refined $\chi = 1$ crossing cells: 4,709.
- Median χ approximately 1.0014.

Focused top-two lobe refinement:

- Tier-2 crossing cells: 12,061.
- Median χ approximately 0.9987.

Focused tier-3 refinement:

- Tier-3 crossing cells: 12,955.
- Median χ approximately 0.9998.

These results do not certify topology. They do show that the candidate $\chi = 1$ structures did not evaporate under focused refinement.

Dense Node-B radial audit:

- Rays: 1,024.
- Radial samples per ray: 192.
- Total $\chi = 1$ crossings: 1,032.
- No-crossing rays: 509.
- Single-crossing rays: 40.
- Multi-crossing rays: 475.
- Maximum crossings on one ray: 4.
- Median crossing radius: approximately 0.9847.

This radial result confirmed that the BCW object was not a simple star-shaped shell around Node B. More rays filled the visual structure more cleanly, but did not collapse the geometry into a single crossing per direction.

The provisional lesson is that the barycenter compression residual is operationally real in the sandbox. The $\chi = 1$ structure persists under focused refinement. Cartesian and radial witnesses agree that the object is multi-lobed. Topology remains uncertified.

16 Interpretation: What Static Context Can Mean

On a static time slice, BCW approaches the parent-context problem by separating declaration, compression, and residual testing.

A parent context declares where the child belongs. A child/group declaration says what is resolved. A surrogate says how the child may be compressed for the parent. The BCW residual says where that compression fails. The ledger records the result.

This does not solve dynamics. It does not answer how context updates across time, how influence propagates causally, how worldlines carry identity, or whether BCW supports trajectory prediction. Those require temporal ledgers, moving scenes, retarded or foliation-aware declarations, and geodesic or congruence diagnostics.

But on a time slice, the operational context grammar is clear:

- resolved child structure must be carried where compression fails;
- parent-readable surrogates may be used where compression succeeds;
- contested seams require overlap, refinement, and retention discipline.

This is the static-slice prize. It is not a magic context boundary. It is a bookkeeping calculus.

17 Limits

BCW-L001 is synthetic. It is not astrophysical evidence.

BCW-L001 used only the $L = 0$ barycenter monopole surrogate.

BCW-L001 did not include an explicit external parent field Π .

BCW-L001 did not certify topology.

BCW-L001 did not test grouping sensitivity.

BCW-L001 did not test Node 0 registration sensitivity.

BCW-L001 did not sweep barycenter surrogate softening β_B .

BCW-L001 did not sweep ϵ_χ .

BCW-L001 did not implement the quadrupole surrogate rung.

BCW-L001 did not compare BCW structures to trajectory or geodesic-congruence response.

The method itself is branch-dependent. It is defined inside a declared weak-field electric-Weyl/tidal branch. Stronger GR-native applications will require explicit metric, gauge, observer, foliation, and tensor declarations.

No claim in this paper should be read as a new force, a physical boundary, a causal horizon, a membrane, an orbital stability limit, or a modification of GR.

18 Next Work

The immediate next method target is BCW-L002: a declared child-in-parent synthetic scene.

BCW-L002 should include an explicit parent context Π , declared child group G , resolved child field $E_{\text{actual},G}$, surrogate field $E_{\text{sur},G}^{(0)}$, parent field E_Π , self-compression witness $\chi_G^{(0)}$, parent-weighted

witness $\chi_{G|\Pi}^{(0)}$, shell-local Cartesian cell complex, Node-B radial audit toggle, Node 0 registration audit, barycenter softening sensitivity, ϵ_χ sensitivity, and grouping sensitivity.

The next surrogate-ladder target is $L = 2$, introducing a quadrupole-augmented surrogate only after its group-declaration-relative tensor convention is specified.

The next paper target is to turn this review draft into a harvestable internal package with selected figures and the BCW-L001 run note.

19 Conclusion

GFRO was a useful false door. It revealed that Atlas needed field-derived source-context residuals, but it over-promoted residual emission into a full context solution.

BCW is the corrected doorframe.

It asks a narrower, stronger question: where does a resolved group fail to compress into its declared parent-readable surrogate?

The answer is not emitted by pairwise equality. It is obtained through declared source rosters, surrogate rules, tensor residuals, datum support, ledgers, refinement, and audit.

Context, on a static slice, is not discovered as a pre-existing shell. It is declared, compressed, tested, ledgered, and audited.

A Datum Bible Insert

Minimum included rules:

- Surface is object; band is witness support.
- Scalar Q may scaffold; BCW tensors adjudicate group-context compression.
- Node B is a compression witness, not a source.
- Node G is a morphology witness, not a source.
- Cartesian cells must remain globally aligned.
- Radial audit is toggleable witness mode, not primary extractor by default.
- Rendering audits but never decides.
- ϵ_i , ϵ_B , and ϵ_χ are distinct declarations.
- Topology is not certified by render persistence alone.

Expanded Datum Bible integration should cite:

`Atlas_Datum.Bible_v0.3.1_BCW.L001.txt`

B BCW-L001 Run Note Requirements

BCW-L001 is now supported by:

`BCW_L001_synthetic_12_source_run_note_v0.2_exact_metadata_fill.txt`

Remaining before public use:

- confirm ϵ_χ from original first-pass generator or formally declare reproduced evaluator value;
- select exact figure plates;
- run ϵ_B/ϵ_χ sensitivity audits;
- run Cartesian/radial reconciliation;
- run topology certification pass if topology will be claimed.

C Public Glossary

BCW

Barycentric Compression Witnessing. Datum-ledger method for testing where resolved group structure fails to compress into a declared parent-readable surrogate.

Compression residual

Tensor difference between the resolved group field and surrogate field.

Compression-parity surface

Candidate surface where the compression residual norm equals the surrogate norm.

Datum

A registered interrogator, sampler, witness, or reporting structure. A datum does not contribute source terms.

ϵ_χ Numerical denominator floor in χ . Not a physical scale.

ϵ_B Barycenter surrogate softening. Protects a non-physical comparison model.

ϵ_i Source-model softening associated with a physical source contribution model.

Node B

Mass barycenter of a declared group, used as compression witness. Not a source.

Node G

Geometric centroid of a declared group, used as morphology witness. Not a source.

Pairwise scaffold

Legacy scalar equality branch that drafts candidate seams but does not solve group-context compression.

Surrogate ladder

Sequence of parent-readable approximations to a resolved group, beginning with the barycenter monopole and potentially adding declared multipole rungs.

Surface/support distinction

The BCW surface is the mathematical $\chi = 1$ set. The cell/radial band is the finite witness support used to sample and audit it.

Topology certification

Evidence that connectivity, holes, tunnels, and component structure converge under declared reconstruction and sensitivity tests.

D Final Worklist Before Public or Repo Promotion

Required:

1. Confirm or declare ϵ_χ .
2. Select/regenerate figure plates.
3. Attach the BCW-L001 run note.
4. Decide whether to keep the paper in `.txt` or typeset to PDF.
5. Harvest to repo only after Daniel authorizes.

Strongly recommended:

1. Run ϵ_B and ϵ_χ sensitivity audits.
2. Run BCW-L002 before any public claim stronger than “method proposal.”
3. Get one more Claude/Codex review after figure selection.

References

- [1] Scieszinski, D. *The Gravitational Coherence Surface: Parity Horizons of Gravitational Identity*. April 2026.
- [2] Scieszinski, D. *Many-Source Parity Networks in Weak-Field Gravitational Readability*. April 2026.
- [3] Scieszinski, D. *Mesoscale Gravitational Readability: Derived Interface Geometry Between Local Evaluation and Coarse Averaging*. April 2026.
- [4] Scieszinski, D. *Proof-of-Concept Extraction of Many-Source Gravitational Readability Structure*. April 2026.

- [5] Scieszinski, D. *The Nesting Domain Handoff Problem: Parity Networks and the Modular Structure of Gravitational Scenes*. April 27, 2026.
- [6] Scieszinski, D. *Gravitational Coherence Surfaces as Information Boundaries of Source Attribution*. Draft v0.2, May 2026.
- [7] Scieszinski, D. *Three Measures of Gravitational Identity*. 2026.
- [8] Scieszinski, D. *Atlas Datum Bible v0.3.1 BCW-L001*. Internal method note, May 2026.
- [9] Buchert, T. “On average properties of inhomogeneous cosmologies.” *General Relativity and Gravitation* 32, 105–125 (2000).
- [10] Geroch, R. “Multipole moments. II. Curved space.” *Journal of Mathematical Physics* 11, 2580–2588 (1970).
- [11] Hansen, R. “Multipole moments of stationary space-times.” *Journal of Mathematical Physics* 15, 46–52 (1974).
- [12] Loffler, F. et al. “The Einstein Toolkit: A community computational infrastructure for relativistic astrophysics.” *Classical and Quantum Gravity* 29, 115001 (2012).
- [13] Wald, R. M. *General Relativity*. University of Chicago Press, 1984.
- [14] Shannon, C. E. “A Mathematical Theory of Communication.” *Bell System Technical Journal* 27, 379–423 and 623–656 (1948).